Workforce Analytics | Technical Report

Protocol

This is an educational study, never an employee decision system. The fictional HR sample answers workforce questions; Adult independently audits historical income classification. No row joins or category equivalences connect these populations. BLS is contextual only.

1. Validate raw hashes, schemas, labels, missingness, uniqueness and ranges. Retain exact original files locally, excluded from Git. Impute within fitted pipelines. Drop HR identifiers/constants and Gender; keep Gender alongside outcomes for subgroup reporting. Exclude Adult sex/race and sampling weight from predictors. Proxies remain, so omission does not guarantee fairness.
2. HR uses seeded stratified 60/20/20 train/validation/test. Compare logistic regression and random forest on validation average precision; optimize selected model threshold on validation F1. No refitting after threshold selection. Adult keeps official test file untouched; split training file 80/20 and apply same selection protocol. Report test metrics only after choices are frozen. This evaluates cross-sectional classification, not future exits or causal effects.
3. Report precision, recall, F1, ROC AUC and average precision (AP, not trapezoidal PR AUC), confusion counts, Brier scores and calibration bins. No accuracy-only selection. Missing subgroup denominators yield undefined metrics. Wilson intervals quantify binomial uncertainty; subgroup sample sizes and actual/predicted positives are explicit.
4. Fairness compares Female minus Male within each dataset and Female/Male predicted-positive ratios. Adult positive means income above $50K, not a hiring decision. Attrition positive means flagged attrition, which is not a favorable employment outcome. These distinct meanings forbid treating their selection rates as equivalent.
5. Compensation uses unspecified source units. Welch and Mann-Whitney tests are exploratory; role-adjusted log-income regression assesses conditional association. Report effect sizes, uncertainty and confounding limitations. Synthetic patterns cannot establish real wage inequality or discrimination.
6. BLS January 2026 medians use a different, contemporary survey population; show separately without performance traffic lights or claimed matched comparisons. Recommendations concern data collection, workload investigation and controlled pilots, not punitive employee ranking.

SIMULATED = fictional IBM source records. OBSERVED = Adult records or BLS published estimates. DERIVED = calculations, always retaining source status. ASSUMPTION = thresholds, split seed and any business scenario. No real savings are claimed.

Adult official test overlap audit found 23 exact full-record matches and 26 predictor-only matches with the original training file. The official split is preserved; unknown identity and repeated records can make evaluation optimistic. Leakage control describes fitting/selection discipline, not a guarantee of distinct people.

Data validation

Hashes verified before every run: {'emp_attrition.csv': 'a5c31e38bd7fafc9bc333884eb181b06b41b8e5e488e8f7ccb27199fb3be7659', 'adult.zip': '7537312dd56c2b98035880805ce99e68183a30ee468aa5329d6df0fbb3cc21bb'}

HR dimensions 1470 x 35; missing cells 0. Exact duplicates 0. IQR flags are diagnostic only.

Adult original training/test sizes 32561/16281; missing training counts {'age': 0, 'workclass': 1836, 'fnlwgt': 0, 'education': 0, 'education_num': 0, 'marital_status': 0, 'occupation': 1843, 'relationship': 0, 'race': 0, 'sex': 0, 'capital_gain': 0, 'capital_loss': 0, 'hours_per_week': 0, 'native_country': 583, 'income': 0}. Original test period not claimed distinct in time. Test rows exactly matching original training records: 23; predictor-only matches: 26. Official split preserved; unknown identity and repeated records may make evaluation optimistic.

Results and interpretation

DERIVED FROM SIMULATED DATA: 1,470 HR records include 237 Attrition Yes labels (16.1%); these are not annual turnover rates.

Highest observed fictional role share: Sales Representative, 33/83 (39.8%). Small group support and role composition matter.

Held-out HR logistic at validation-selected threshold 0.32: precision 0.500, recall 0.511, F1 0.505, AP 0.572, ROC AUC 0.785; test n=294.

Independent Adult income benchmark test n=16,281; random_forest, threshold 0.34, F1 0.693, AP 0.790. Positive means >$50K historical income.

Compensation associations

Female-minus-Male mean MonthlyIncome difference 306.06 source units; Cohen d=0.065, Welch p=0.2218; Mann-Whitney p=0.08842, rank-biserial=0.052.

Adjusted Male-vs-Female log-income coefficient 0.0096; HC3 95% CI -0.0199 to 0.0391; p=0.5253. JobRole + Age + TotalWorkingYears; HC3 robust covariance; exploratory association, not causal

Tests are exploratory and not adjusted for multiple comparisons; synthetic records do not establish real pay discrimination. Role adjustment can itself condition on historically biased variables.

Explainability

Largest test permutation AP decreases: OverTime: 0.207 (SD 0.013); JobSatisfaction: 0.085 (SD 0.014); NumCompaniesWorked: 0.063 (SD 0.017); YearsSinceLastPromotion: 0.052 (SD 0.024); TotalWorkingYears: 0.050 (SD 0.026)

Importance is predictive association, not causal leverage; correlated variables can share or obscure importance. Negative values reflect sampling variability. Test importance is explanatory only and is not fed back to model selection.

BLS context

BLS January 2026 medians: overall 4.1, men 4.3, women 4.0 years. Survey wage/salary population differs from fictional undated sample. See separate context CSV.

Sources and license

Fictional IBM HR data; ODbL 1.0 / DbCL 1.0. Derived educational analysis, not employee decisions.

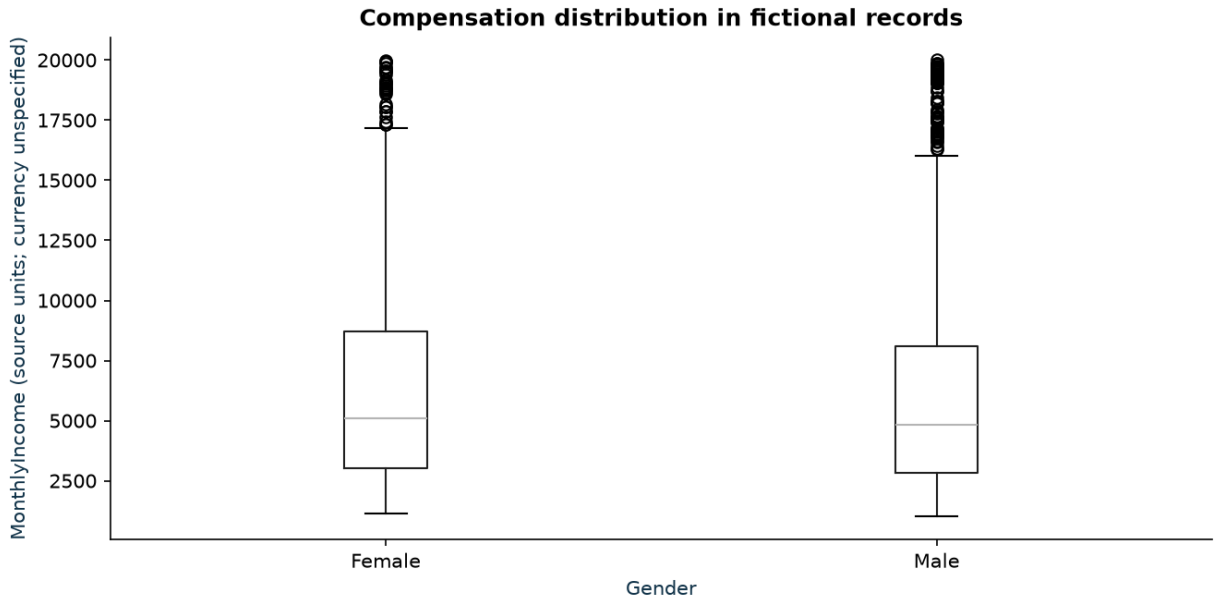
IBM HR source and license: https://github.com/IBM/employee-attrition-aif360#dataset-license-details

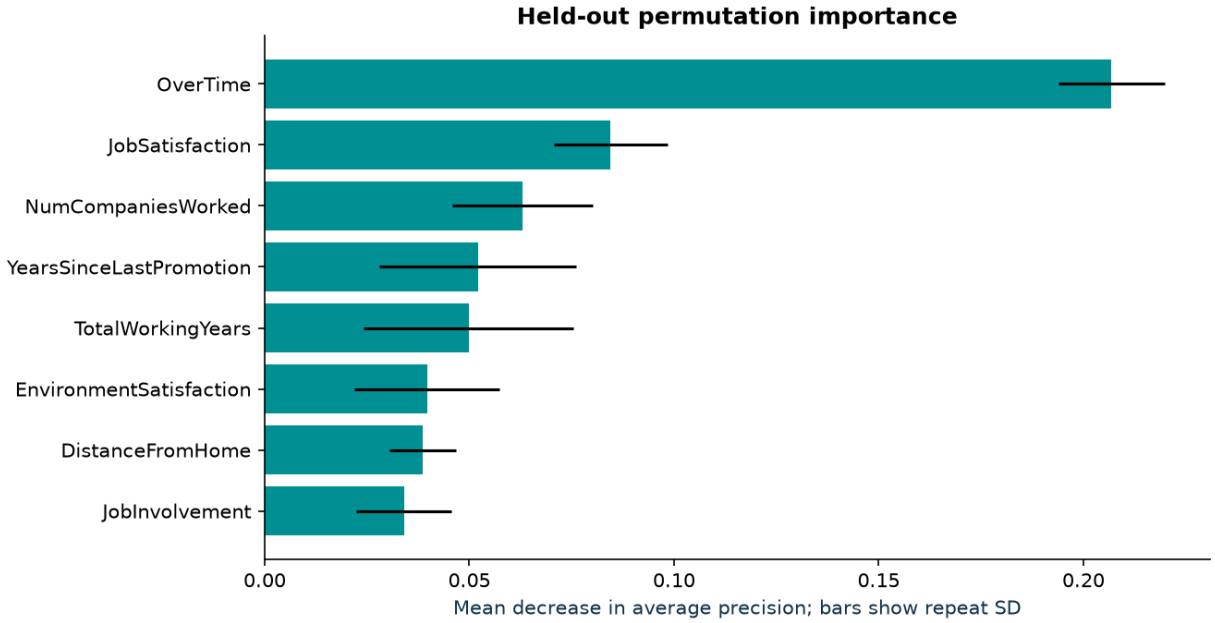
Original fictional HR listing: https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Adult: Becker & Kohavi (1996), DOI 10.24432/C5XW20: https://archive.ics.uci.edu/dataset/2/adult

BLS Employee Tenure in 2026, released September 24 2026: https://www.bls.gov/news.release/tenure.nr0.htm

Data terms: ODbL 1.0 and DbCL 1.0; full license links and alteration methods are in NOTICE.md. Adult: CC BY 4.0. BLS statistics: public domain.





Fictional IBM HR data; ODbL 1.0 / DbCL 1.0. Derived educational analysis, not employee decisions.